

TOPIC 09

FUNCTION (PART 1)

Fundamentals of Computing
(FSPK0022)

Introduction

- What is a function?
- Types of C++ functions:
 - Standard functions
 - User-defined functions
- C++ function structure
 - Function signature
 - Function body
- Declaring and Implementing C++ functions
- Sharing data among functions through function parameters
 - Value parameters

What is Function?

Function Overview

- We use functions (aka: modules, methods, procedures, subprocedures, or subprograms) to perform a specific task. In algebra, a function is defined as a rule or correspondence between values, called the function's arguments, and the unique value of the function associated with the arguments.
- For example:

If **$f(x) = 2x + 5$** , then **$f(1) = 7$** , **$f(2) = 9$** , and **$f(3) = 11$**

You would say the call $f(1)$ returns the value 7

- 1, 2, and 3 are arguments
- 7, 9, and 11 are the resulting values (or corresponding values)

Function Overview (cont.)

C++ functions work in largely the same way. Format of a C++ function call:

functionName(argumentList)

where the **argumentList** is a comma-separated list of arguments (data being sent into the function).

In general, function arguments may be **constants, variables** or more complex **expressions**.

Why Using Function?

- **Divide and Conquer**
 - Allow complicated programs to be divided into manageable components
 - Programmer can focus on just the function: develop it, debug it, and test it
 - Various developers can work on different functions simultaneously
- **Reusability:**
 - Can be used in more than one place in a program--or in different programs
 - Avoids repetition of code, thus simplifying code maintenance
 - Can be called multiple times from anywhere in the program
- **Components:**
 - Custom functions and classes that you write
 - Prepackaged functions and classes available in the C++ Standard Library

Predefined Function

Library Function (Predefined)

- Predefined functions
 - Part of the C++ language
 - Provided in function libraries
- Examples:
`abs(x)` , `sin(x)` , `log(x)` , `pow(x, n)`
- These functions will return a value
 - To be printed `cout << sin(x);`
 - To be assigned `y = pow(3, 4.5);`
 - To be used in an expression `3.14 * sqr(r)`

Library Function (Predefined)

- “Built-in” functions that **come with the compiler**.
- The source code (definition) for library functions does **NOT** appear in your program.
- To use a library function, you simply need to **include the proper header file** and know the name of the function that you wish to use.

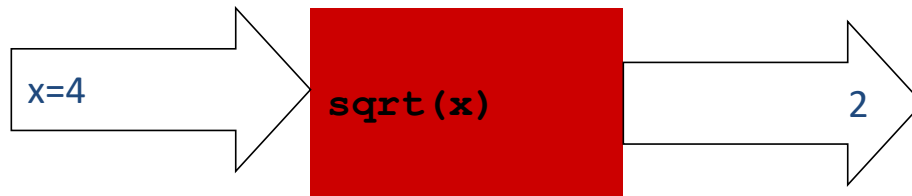
#include compiler directive

Library Function (Predefined)

Compiler directive	Purpose
<code><cctype></code>	Character classification and conversion
<code><cmath></code>	Math functions
<code><stdlib></code>	Data conversion
<code><time></code>	Time functions

Library Function (Predefined)

- A collection of specialized functions.
- C++ promotes code reuse with predefined classes and functions in the standard library
- The functions work as a black box:



Math Functions

- Required header: `#include <cmath>`
- Example functions

Function	Purpose
<code>abs(x)</code>	returns the absolute value of an integer.
<code>pow(x,y)</code>	calculates x to the power of y. If x is negative, y must be an integer. If x is zero, y must be a positive integer.
<code>pow10(x)</code>	calculates 10 to the power of x.
<code>sqrt(x)</code>	calculates the positive square root of x. (x is ≥ 0)

Library Functions: Example 1

```
#include <iostream>
#include <cmath>
using namespace std;
int main()
{
    double area, radius;

    cout<< "This program calculates the area of a
    circle.\n";
    cout<<"What is the radius of the circle? ";
    cin>>radius;
    area=3.14159 * pow(radius,2.0);
    cout<<"The area is " << area <<endl;
    system ("pause");
    return 0;
}
```

Library Functions: Example 2

```
#include <iostream>
#include <cmath>
using namespace std;
main() {
    int nom1,nom2, result;
    cout<<"Enter two numbers";
    cin>>nom1>>nom2;
    if ((nom1<0) || (nom2<0))
    {    cout<<"negative number/s";
    }
    else
    {
        result= sqrt(nom1 + nom2);
        cout<<"The square root of "<< nom1+nom2 << "is"
        << result;}
    }
```

More Mathematical Functions

Function	Standard Library	Purpose: Example	Argument(s)	Result
<code>abs(x)</code>	<code><cstdlib></code>	Returns the absolute value of its integer argument: if <code>x</code> is <code>-5</code> , <code>abs(x)</code> is <code>5</code>	<code>int</code>	<code>int</code>
<code>ceil(x)</code>	<code><cmath></code>	Returns the smallest integral value that is not less than <code>x</code> : if <code>x</code> is <code>45.23</code> , <code>ceil(x)</code> is <code>46.0</code>	<code>double</code>	<code>double</code>
<code>cos(x)</code>	<code><cmath></code>	Returns the cosine of angle <code>x</code> : if <code>x</code> is <code>0.0</code> , <code>cos(x)</code> is <code>1.0</code>	<code>double</code> (radians)	<code>double</code>
<code>exp(x)</code>	<code><cmath></code>	Returns e^x where $e = 2.71828\dots$: if <code>x</code> is <code>1.0</code> , <code>exp(x)</code> is <code>2.71828</code>	<code>double</code>	<code>double</code>
<code>fabs(x)</code>	<code><cmath></code>	Returns the absolute value of its type double argument: if <code>x</code> is <code>-8.432</code> , <code>fabs(x)</code> is <code>8.432</code>	<code>double</code>	<code>double</code>
<code>floor(x)</code>	<code><cmath></code>	Returns the largest integral value that is not greater than <code>x</code> : if <code>x</code> is <code>45.23</code> , <code>floor(x)</code> is <code>45.0</code>	<code>double</code>	<code>double</code>

More Mathematical Functions

<code>log(x)</code>	<code><cmath></code>	Returns the natural logarithm of x for $x > 0.0$: if x is 2.71828, <code>log(x)</code> is 1.0	double	double
<code>log10(x)</code>	<code><cmath></code>	Returns the base-10 logarithm of x for $x > 0.0$: if x is 100.0, <code>log10(x)</code> is 2.0	double	double
<code>pow(x, y)</code>	<code><cmath></code>	Returns x^y . If x is negative, y must be integral: if x is 0.16 and y is 0.5, <code>pow(x, y)</code> is 0.4	double, double	double
<code>sin(x)</code>	<code><cmath></code>	Returns the sine of angle x : if x is 1.5708, <code>sin(x)</code> is 1.0	double (radians)	double
<code>sqrt(x)</code>	<code><cmath></code>	Returns the non-negative square root of x ($\sqrt{x}$) for $x \geq 0.0$: if x is 2.25, <code>sqrt(x)</code> is 1.5	double	double
<code>tan(x)</code>	<code><cmath></code>	Returns the tangent of angle x : if x is 0.0, <code>tan(x)</code> is 0.0	double (radians)	double

More Mathematical Functions

- Require `cmath` header file
- Take `double` as input, return a `double`
- Commonly used functions:

<code>sin</code>	Sine
<code>cos</code>	Cosine
<code>tan</code>	Tangent
<code>sqrt</code>	Square root
<code>log</code>	Natural (e) log
<code>abs</code>	Absolute value (takes and returns an int)